

Command Sequence to run a job through SLURM from git repository.  
You can change the path to SLURM script to available scripts, change highlighted parts.  
Replace YOUR\_SJSU\_ID with your actual ID

Login to HPC: ssh YOUR\_SJSU\_ID@coe-hpc1.sjsu.edu

Password:

Change to hpc3: ssh coe-hpc3

Check hostname: hostname

Check current directory: pwd

Run interactive job:

```
srun \  
--partition=gpuqs \  
--gres=gpu:1 \  
--nodes=1 \  
--ntasks=1 \  
--cpus-per-task=4 \  
--mem=16G \  
--time=00:30:00 \  
--pty /bin/bash
```

```
hostname  
echo "$SLURM_JOB_ID"  
echo "$CUDA_VISIBLE_DEVICES"  
nvidia-smi
```

exit

On your computer:

git clone <https://github.com/sonalilonkar1/carla-simulator.git>

scp -r ~/carla-simulator YOUR\_SJSU\_ID@coe-hpc1.sjsu.edu:~/carla-test-demo

On HPC g17 node:

cd carla-test-demo

Create output directories(One time):

```
mkdir -p /scratch/cmpe281-fa26/$USER/logs
```

```
mkdir -p /scratch/cmpe281-fa26/$USER/output
```

Submit the job through SLURM:

```
JOB_ID=$(sbatch --parsable lab3/accident_report/run_accident_report.sbatch)
```

echo "\$JOB\_ID"

Monitor it:

squeue -j "\$JOB\_ID"

Check logs and results:

cat /scratch/cmpe281-fa26/\$USER/logs/lab3-accident-\$JOB\_ID.out

cat /scratch/cmpe281-fa26/\$USER/logs/lab3-accident-\$JOB\_ID.err

Set the result directory:

OUTPUT\_DIR="/scratch/cmpe281-fa26/\$USER/output/accident\_report/\$JOB\_ID"

Inspect the results:

find "\$OUTPUT\_DIR" -maxdepth 2 -type f | sort

cat "\$OUTPUT\_DIR/result.json"

cat "\$OUTPUT\_DIR/incident\_report.json"

head "\$OUTPUT\_DIR/telemetry.csv"

In your Machine:

Make directory in your machine for results:

JOB\_ID=**add job id here**

MAC\_DIR="\$HOME/**Your-Path-to-directory**/carla-lab3-accident-\$JOB\_ID"

mkdir -p "\$MAC\_DIR"

(Always run this from your machine)

Copy output files from HPC to your machine:

scp -r

"018228067@coe-hpc1.sjsu.edu:/scratch/cmpe281-fa26/018228067/output/accident\_report/\$JOB\_ID/." "\$MAC\_DIR/"

Go to your machine directory: cd **Your-Path-to-directory**/carla-lab3-\$JOB\_ID

Convert camera frames to video:

ffmpeg \

-y \

-framerate 20 \

-i 'frames/frame-%06d.png' \

-c:v libx264 \

-pix\_fmt yuv420p \

"lab3-accident-\$JOB\_ID.mp4"

Open the video: open "lab3-accident-\$JOB\_ID.mp4"